The estimator, not the geography: scaffold attacks hide a racial price penalty from LIME, KernelSHAP and GeoShapley

Draft v1 for discussion with Peng Luo

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Abstract

Post-hoc explanations are increasingly used to audit machine-learning models for spatial bias, and GeoShapley was designed to separate location effects from other features. We show that a scaffold attack can hide a racial price penalty from LIME, KernelSHAP and GeoShapley. A home-valuation model for King County, Washington, halves predicted prices in census tracts where Black and Latino residents exceed a quarter of the population; a detector trained on each explainer's own queries serves an innocuous model to the explainer and the biased model to real homes. LIME then shows 0.02 of the penalty while 97.4% of the penalised homes still pay it. KernelSHAP and GeoShapley show 0.67 and 0.72 of it at about 90% fidelity; the trigger drops in their global importance ranking but stays prominent on the penalised homes, and the attacker can hide more only by withholding the penalty from real homes. Treating location as a joint player protects GeoShapley little: it keeps at most a few points more of the penalty than KernelSHAP, and under attack its location effect shrinks while spurious feature-by-location interactions appear. The difference between explainers is set by their estimators, not by their treatment of space: efficiency keeps the penalty inside a Shapley explanation, and the share kept is close to the average, over coalition sizes, of the share of hybrids the detector routes to the biased model. The results are the same with random forests and neural networks, because the explainers' queries do not depend on the model, and survive changes to the Shapley background; only switching LIME to a sampler the detector was not trained on recovers part of the penalty. A residual check that needs no explainer separates the penalised homes from the rest (AUC 0.81 for a 21% penalty).

1 Introduction

Machine-learning models now set or inform home valuations, insurance prices and lending decisions, and in each of these location is among the strongest predictors. Location is also a proxy for race and ethnicity, so a valuation model can reproduce the effects of historical redlining without ever being given a demographic variable [CITE: redlining and algorithmic valuation]. Regulators, lenders and researchers increasingly check such models with post-hoc explanation methods, which attribute each prediction to the model's inputs.

Geographers have adapted these methods to spatial data. GeoShapley (Li, 2024) treats the coordinate pair as one player in a Shapley game and separates an intrinsic location effect from location-invariant feature effects and feature-by-location interactions. For a fairness audit this is attractive: if a model penalises a neighbourhood, the location effect should show it.

That reading assumes the model cooperates. Slack et al. (2020) showed that LIME and SHAP can be deceived by a *scaffold*: a model that recognises the synthetic inputs an explainer generates and answers them with an innocuous model, while real inputs receive biased predictions. The attack has been shown only on non-spatial tabular benchmarks. Whether an explainer built for spatial data inherits the weakness, resists it, or fails in its own way has not been examined.

We examine it on a home-valuation model for King County, Washington. The model halves predicted prices in census tracts where Black and Latino residents exceed a quarter of the population, and a scaffold hides the penalty behind a detector trained on each explainer's own queries. Our central finding is that *how much a scaffold can hide is set by the explainer's estimator — where it queries the model — and not by whether the explainer knows about geography.* LIME samples far from real homes and can be fooled almost for free. KernelSHAP and GeoShapley satisfy efficiency and evaluate the explained home itself, together with many near-real hybrids, so the attacker can hide the penalty from them only by also withholding it from real homes. We measure that stealth–fidelity trade-off and explain it with an argument about Shapley coalitions that accounts for the measured trade-off.

We ask five questions.

1. Can a scaffold hide a demographic price penalty from LIME, KernelSHAP and GeoShapley in a real housing market?
2. What does hiding cost the attacker, in the share of real homes that still receive the bias?
3. Does GeoShapley's treatment of location make it harder to fool than a non-spatial Shapley explainer, and what happens to its decomposition under attack?
4. Why do the explainers differ?
5. Do the answers survive other model families, auditors who change the explainer's settings, a smaller penalty and weaker spatial autocorrelation of the biased attribute?

In brief: all three explainers can be fooled. LIME is fooled completely, keeping 0.02 of the penalty while the scaffold still charges 97.4% of the penalised homes; an earlier report that LIME resists the attack in a geographic setting came from a detector that did not match LIME's sampler. KernelSHAP and GeoShapley keep 0.67 and 0.72 of the penalty when the attacker serves the biased model to about 90% of real homes (85.9% and 87.9% of the penalised ones), and the share they keep falls roughly in step with the share of homes the attacker is willing to spare. On the penalised homes themselves the trigger stays among the three largest attributions; the scaffold shrinks it and pushes it down the explainer's global importance ranking, but cannot take it out of those homes' explanations. Treating location as a joint player protects GeoShapley little: in the feature design GeoShapley keeps slightly more of the penalty than KernelSHAP, in the location design no detectable difference between GeoShapley and KernelSHAP, and under attack its location effect shrinks while its feature-by-location interactions roughly double. The share a Shapley explainer keeps is close to the average, over coalition sizes, of the share of its hybrids that the detector routes to the biased model. These results hold with random forests and neural networks and when the auditor changes the Shapley background; only an auditor who switches LIME to a sampler the attacker did not anticipate recovers part of the penalty. Finally, a residual check that uses no explainer separates the penalised homes from the rest.

2 Background

Perturbation-based explanation, and where each method queries the model. All three explainers estimate attributions by evaluating the model at inputs that are not real observations; this is the attack surface.

- **LIME** (Ribeiro et al., 2016) fits a weighted linear surrogate around an instance. Its default tabular sampler discretises each feature into training quartiles and draws new rows bin by bin, so almost every query is a combination of feature values that no real home has. The instance itself is the first row.
- **KernelSHAP** (Lundberg and Lee, 2017) estimates Shapley values by evaluating coalitions: the features in a coalition keep the instance's values and the rest are taken from a background row. With M features it needs $2^M - 2$ coalitions per background row (2,046 in our feature design, 1,022 in the location design), which it enumerates in full.
- **GeoShapley** (Li, 2024) uses the same coalition construction but moves the two coordinates together as one player, so every hybrid sits at a real location. It reports an intrinsic location effect (GEO), location-invariant primary effects and feature-by-location interaction effects that together satisfy efficiency.

The scaffold attack. Slack et al. (2020) train a classifier d to tell real inputs from an explainer's synthetic queries and deploy

$$e(x) = \begin{cases} f(x) & d(x) \geq \tau \quad \text{(looks real: biased answer)} \\ \psi(x) & d(x) < \tau \quad \text{(looks synthetic: innocuous answer).} \end{cases} \tag{1}$$

Real users receive biased predictions; the explainer, querying mostly synthetic points, reports the innocuous model. The attack is only as good as the detector's match to the explainer's sampler, which is why Slack et al. (2020) train one detector per explainer.

Related work. Explanations can be manipulated by modifying the model [CITE: Heo et al. 2019; Dombrowski et al. 2019], and rationalised to look fair [CITE: Aïvodji et al. 2019, fairwashing]. Defences detect or avoid off-manifold queries (Carmichael and Scheirer, 2023) [CITE: on-manifold Shapley, Frye et al.]. Shapley values in geography (Li, 2022) and GeoShapley (Li, 2024) are increasingly used to read spatial effects out of machine-learning models. [**TODO:** position against the geospatial XAI and algorithmic-bias literature Peng wants cited; every entry to be checked against the published version]

3 Study area and data

Housing. 3,000 house sales in King County, Washington, from the sample distributed with the GeoShapley package (`seattle_sample_3k.csv`) [**TODO:** confirm the sale years of the sample]. Each sale has eight structural features (bathrooms, living area, lot area, grade, condition, waterfront, view, age), projected coordinates (UTM zone 10N, metres) and the base-10 logarithm of the sale price (median sale $435,000).

Demographics. American Community Survey 2011–2015 table B03002 at census-tract level for the 398 King County tracts, joined to 2015 TIGER/Line tract polygons. A point-in-polygon join

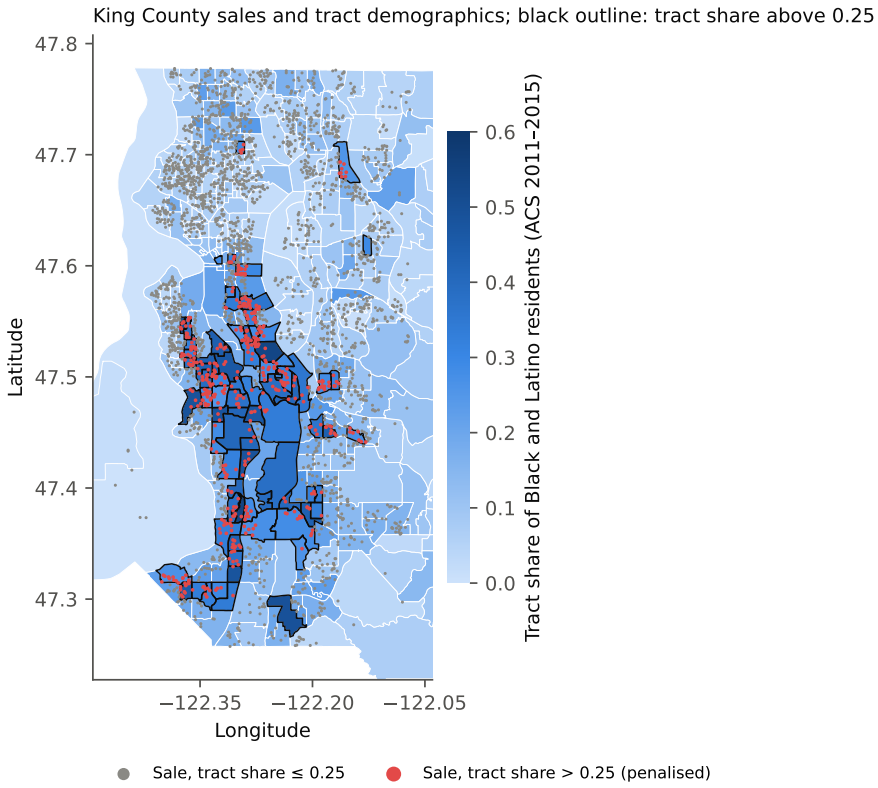


Figure 1: Study area. Tracts shaded by the share of Black and Latino residents; tracts above the 0.25 threshold outlined; sales shown as points, red where the biased model applies its penalty.

assigns every sale to one of 334 tracts and attaches the tract share of Black (non-Hispanic) and Hispanic or Latino residents, `pct_black_latino`. Figure 1 maps the share and the sales.

Split. For each seed s the 3,000 sales are split 80/20 into 2,400 training and 600 test homes. The first 200 test homes are the homes explained; the other 400 calibrate the attacker's threshold (Section 4.3).

Spatial structure. The trigger is strongly autocorrelated: Moran's I of `pct_black_latino` on an 8-nearest-neighbour, row-standardised weights matrix is 0.83.

4 Methods

4.1 Honest, biased and innocuous models

The honest model h is a gradient-boosted regressor (XGBoost, 500 trees, depth 6, learning rate 0.1) on the eight structural features and the two coordinates. The biased model subtracts a fixed penalty δ for homes in tracts where the trigger exceeds 0.25:

$$f(x) = h(x) - \delta\,\mathbf{1}\{T(x) > 0.25\}. \tag{2}$$

Design	Seeds	Honest R^2	Innocuous R^2	Test homes penalised (%)	Penalised among 200 explained	Raster = tract file (%)
Feature	5	0.87 ± 0.01	0.65 ± 0.04	15.9 ± 1.5	32.2 ± 6.5	–
Location	5	0.87 ± 0.01	0.65 ± 0.04	15.9 ± 1.4	32.0 ± 6.1	100.0 ± 0.1

Table 1: Data and models per design, mean ± 95% interval over seeds. R^2 on the 600 test homes. The raster column checks that the model's coordinate lookup puts each test home in the same tract as the spatial join in the data file.

With the price on a base-10 log scale, $\delta = 0.30$ multiplies the predicted price by $10^{-0.3} \approx 0.50$: a penalty of one half.[1] The innocuous model ψ is a second XGBoost regressor (300 trees, depth 5) on the eight structural features only, so any explanation of it is free of location and demographics. Table 1 gives the fit of both on the test homes.

Two designs. In the *feature design* the trigger $T(x)$ is the `pct_black_latino` column, which the scaffold sees as an eleventh feature. In the *location design* there is no demographic column: the model sees the ten real features, and $T(x)$ is the tract share looked up from the home's coordinates inside the model (a 25 m raster of the real tract polygons). This is redlining in the literal sense, and it is the case GeoShapley's location effect exists to expose. A third design drops the coordinates altogether (LIME and KernelSHAP only).

4.2 The scaffold and its detectors

The deployed model is (1) with a 100-tree random forest as d. We build d two ways.

- **Generic detector**: trained on the perturbation recipes of Slack et al. (2020): real rows plus Gaussian noise for LIME, and real rows with features swapped in from ten k-means centroids for the Shapley explainers. Threshold $\tau = 0.5$.
- **Matched detector**: trained on the queries each explainer actually sends. For LIME the negatives are draws from `LimeTabularExplainer`'s own sampler around every training home. For KernelSHAP and GeoShapley we run the explainer on training homes with a background drawn differently from the auditor's, record a random subset of every row it submits, and train d to separate them from real training rows. Rows identical to a real home (the instance itself) are labelled real.

Detectors never see the test homes.

4.3 Thresholds and fidelity

An attacker must decide how many real homes it is willing to misroute to ψ. We set the matched threshold on the 400 calibration homes: $\tau^* = \min(0.5, q_{0.10})$, where $q_{0.10}$ is the 10th percentile of their detector scores, so at least 90% of unseen real homes receive f. *Fidelity* is then measured on the 200 explained homes, which did not set τ^*. Because $f = h$ on unpenalised homes, the bias actually delivered is the fidelity on the penalised explained homes, which we report alongside. To trace the frontier we lower fidelity on purpose with the same detector, at thresholds giving 75% and 50% fidelity on the calibration homes.

[1] Earlier project documents gave this as a 26% penalty, which assumes natural logarithms; Appendix D and the repository's CHANGELOG list every correction of this kind.

4.4 Explainers

LIME with 500 samples and the default quartile sampler; KernelSHAP with a 100-row background sample, every coalition enumerated and no feature selection (`l1_reg=False`, so the values are exact Shapley values for that background)[2]; GeoShapley with the same 100-row background and the coordinates as the joint player. GeoShapley is exact in the location design (eight non-location features); with nine non-location features (feature design) the package evaluates all coalitions of size at most three and their complements plus Shapley-weighted draws, about 600 of 1,024, which we seed per run.

4.5 What we measure

For explained home i with attribution vector $\phi(x_i)$, trigger set J (the `pct_black_latino` column; both UTM columns for LIME and KernelSHAP in the location design; the location player for GeoShapley in the location design) and penalised set $P = \{i : T(x_i) > 0.25\}$. GeoShapley's attributions enter in the package's Shapley-equivalent form, in which each feature-by-location interaction is split equally between the feature and the location player; the location player therefore carries the intrinsic GEO effect plus half of every interaction, and Table 5 reports the GEO effect and the interactions separately. The measures are:

$$\text{signed attribution} \quad a = \frac{1}{|P|} \sum_{i \in P} \sum_{j \in J} \phi_j(x_i), \tag{3}$$

$$\text{penalty share kept} \quad K = a^{\text{deployed}} / a^{\text{biased}}, \tag{4}$$

$$\text{net of honest (location design)} \quad K_{\text{net}} = (a^{\text{deployed}} - a^{\text{honest}}) / (a^{\text{biased}} - a^{\text{honest}}), \tag{5}$$

$$\text{trigger in top three} \quad \tfrac{1}{|A|} \textstyle\sum_{i \in A} \mathbf{1}\{\text{rank of } \sum_{j \in J} |\phi_j(x_i)| \leq 3\}, \tag{6}$$

$$\text{fidelity} \quad \tfrac{1}{|A|} \textstyle\sum_{i \in A} \mathbf{1}\{d(x_i) \geq \tau\}, \tag{7}$$

$$\text{queries reaching } f \quad \text{share of the explainer's own queries with } d \geq \tau, \tag{8}$$

$$\text{detector AUC} \quad \Pr\{d(x_{\text{real}}) > d(x_{\text{query}})\} + \tfrac{1}{2} \Pr\{=\}. \tag{9}$$

where A is either all $N = 200$ explained homes or the penalised ones, P; we report both. In the location design the trigger score of LIME and KernelSHAP is the sum over both coordinates, ranked against single features, so their top-three shares there are not directly comparable with GeoShapley's single location player. The reference in (4) is the explanation of the biased model served directly, with no scaffold: that, not the honest model, is what the attack has to hide. $K = 1$ means the penalty is fully visible, $K = 0$ fully hidden. Every detector score of every query is histogrammed, so AUC and the leak at any fidelity are computed exactly without re-running an explainer.

4.6 Why Shapley explainers resist: efficiency and coalition sizes

For a Shapley explainer with background B the attributions of a home sum to the served prediction there minus its background mean (efficiency):

$$\sum_{j=1}^{M} \phi_j(x) = e(x) - \frac{1}{|B|} \sum_{b \in B} e(b). \tag{10}$$

[2]shap's default `l1_reg="num_features(10)"` keeps at most ten features per home even under full enumeration, which with eleven features zeroes one of them. At seed 42 switching it off changed the trigger's attribution by less than 0.001.

While the scaffold keeps fidelity, the explained home itself is routed to f and the penalty sits in $e(x)$: it cannot vanish from the explanation; it can only move to other features, or the attacker must stop serving f at x. LIME has no such constraint. The trigger's attribution is an average over coalition sizes with equal weight $1/M$,

$$\phi_j(x) = \frac{1}{M} \sum_{k=0}^{M-1} \mathbb{E}_{|S|=k,\, j\notin S,\, b\sim B} \big[e(x_{S\cup j}, b) - e(x_S, b) \big], \tag{11}$$

where (x_S, b) takes the players in S from x and the rest from b. The largest coalition is the explained home; near-complete coalitions differ from it in one or two players, and near-empty ones are a background (real) home with one or two values swapped in. Both ends look real to a detector. If ρ_k is the share of size-k hybrids routed to f, the share of the penalty kept is roughly the average of ρ_k over sizes, and the attacker can lower it only by rejecting near-real points, which also rejects real homes. We estimate each term of (11) by Monte Carlo with the attack's own detector, background and thresholds (200 hybrids per home and size, drawn independently of the attack run) and compare the result with the measured K (rule M below). Because this evaluates the same Shapley average with the same detector, agreement confirms the size decomposition and our implementation rather than testing a separate hypothesis. The substantive approximation is the plain average over sizes of ρ_k, which ignores how the penalty is spread across sizes; we report it as a description that was not specified in advance.

4.7 Robustness

- **Model family**: h and ψ as random forests (100 trees, leaf size 3) or neural networks (two hidden layers of 256 and 128 units, standardised inputs).
- **Audit settings the attacker did not train for**: Shapley background of 25 or 200 rows, or 100 k-means centroids; LIME with 5,000 samples or with its Gaussian sampler. The detector stays the one trained for the defaults; the biased reference uses the auditor's changed settings.
- **Spatial autocorrelation**: a share of homes receives the trigger value of a random other home, which keeps the marginal distribution and lowers Moran's I.
- **Smaller penalty**: $\delta = 0.10$ (a 21% price cut).

4.8 An outcome-based check

No explainer is involved: on the 600 held-out homes we compute residuals $r = e(x) - y$ of the deployed model against observed log price and ask whether they reveal the penalised homes, using the residual itself, its local mean over the 8 nearest neighbours, global Moran's I, and local Moran (LISA) low–low clusters (Moran, 1950; Anselin, 1995), for penalties from 0.02 to 0.30. The scaffold here is the LIME-matched one, which serves almost every real home with f.

4.9 Statistics and pre-specified decision rules

The seed is the unit of replication; a seed re-runs the split, the models, the detectors, the background draw and the LIME sampling. Each metric is a mean over seeds with a 95% interval $\bar{x} \pm t_{S-1}^{-1}(0.975)\, \hat{\sigma}/\sqrt{S}$; contrasts are paired by seed. The rules below were fixed before the runs.

- **C1 (fooled)**: $1 - K$ under the matched scaffold has an interval above 0, Holm-corrected over the three explainers.

- **C1′ (fully fooled)**: trigger in the top three for at most 5% of the explained homes and $K \leq 0.10$ at fidelity at least 90%; otherwise *partly fooled*. (Computed on penalised homes only, the top-three part gives the same classes.)
- **C2**: GeoShapley minus KernelSHAP in K, matched scaffold; an interval above 0 reads as "GeoShapley slightly harder to fool", an interval containing 0 as no detectable difference at this sample size.
- **C3**: GeoShapley's GEO effect and interaction sums, biased against matched; a change is claimed only if the paired interval excludes 0.
- **C5**: slope of the matched leak on Moran's I over at least three levels.
- **M**: the Monte Carlo prediction holds if its mean absolute error against measured K is no larger than the seed-to-seed standard deviation of measured K; otherwise only the efficiency identity is claimed.
- **R**: a robustness configuration agrees with the main result if each explainer's C1′ class is unchanged.
- **E7**: the outcome-check section is included if, at $\delta = 0.10$, the scaffold's local-mean AUC is at least 0.80 and the honest model's LISA flag rate on unpenalised homes is at most 6%.

5 Results

5.1 Models and detectors

The honest model explains the test prices well (R^2 0.87 in the feature design, 0.87 in the location design) and the innocuous model, which sees neither location nor demographics, less well (0.65 and 0.65; Table 1). The biased model penalises 15.9% of test homes, about 32 of the 200 explained homes in each seed. In the location design the model's coordinate lookup puts 99.97% of test homes in the same tract as the spatial join in the data file.

The matched detectors separate LIME's queries from real homes completely (AUC 1.00), but separate the Shapley explainers' queries only partly (AUC 0.78 for KernelSHAP and 0.72 for GeoShapley in the feature design). This difference, not the attack recipe, drives everything that follows.

5.2 A scaffold hides the penalty from all three explainers

Feature design. Served directly, the biased model's penalty is plain in every explanation: the trigger is among the three largest attributions for every penalised home under all three explainers, and for 67.3% (LIME), 86.3% (KernelSHAP) and 86.6% (GeoShapley) of all explained homes, with a signed attribution on penalised homes of about -0.24 in $\log_{10}$ price (Table 2).

The generic detector of Slack et al. (2020) does not fool LIME here: it lets 95.8% of LIME's queries through to f, and LIME keeps 0.96 of the penalty. This is the source of the earlier project finding that LIME resists the attack in a geographic setting (Appendix D); it is a failure of the detector, not a property of LIME. With a detector matched to LIME's own sampler, no LIME query reaches f (0.0%), the scaffold still serves f to 95.9% of the explained homes and 97.4% of the penalised ones, and LIME keeps 0.02 of the penalty, with the trigger in the top three for 0.0% of penalised homes. LIME is fully fooled.

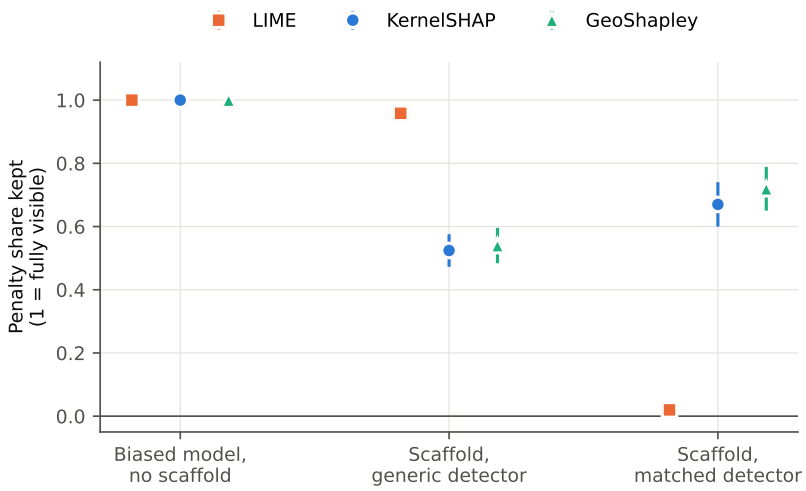


Figure 2: Penalty share kept in the feature design (1 = the explanation shows the whole penalty, as it does when the biased model is served directly; 0 = none of it). Mean and 95% interval over seeds.

Against the Shapley explainers the matched scaffold, at 89.5% and 89.7% fidelity (85.9% and 87.9% on penalised homes), cuts the penalty to 0.67 (95% interval 0.60–0.74) for KernelSHAP and 0.72 (0.65–0.79) for GeoShapley. What it hides is the size of the penalty and its place in the overall ranking, not its presence on the homes that pay it. On penalised homes the trigger stays among the three largest attributions for 99.5% (KernelSHAP) and 100.0% (GeoShapley) of homes. Across all explained homes it is in the top three for 51.7% and 56.6%, against 86.3% and 86.6% without a scaffold, and its rank in mean absolute attribution falls from 2.0 to 3.6 (KernelSHAP) and from 2.0 to 2.8 (GeoShapley). The drop in the all-home top-three share comes from the unpenalised homes, where the trigger's small positive attribution (the model's "bonus" for not being penalised, relative to the background) shrinks under the scaffold: there the trigger is in the top three for 42.3% and 48.1% of homes, against 83.6% and 84.0% without it. Rule C1 (Holm-corrected) gives LIME fooled, KernelSHAP fooled and GeoShapley fooled; by rule C1′ KernelSHAP is partly fooled and GeoShapley partly fooled. The detector cannot do better because 53.6% of KernelSHAP's and 60.8% of GeoShapley's queries look real enough to be served by f.

Location design. Here location has a legitimate price effect, so even the honest model's explanation puts a negative location attribution on the homes the biased model penalises (−0.12 for KernelSHAP). Table 3 therefore gives the share kept both raw and net of the honest model. The LIME-matched scaffold removes location from LIME's explanation altogether: the raw share is 0.05, location is in the top three for 1.6% of penalised homes against 68.6% under the honest model, and the net share is negative (−1.24): LIME now shows less location effect than an honest model would. KernelSHAP and GeoShapley keep 0.48 and 0.48 of the penalty net of the honest model (raw 0.65 and 0.65), at fidelity 90.0% and 88.8% (84.7% and 82.1% on penalised homes). Location stays in their top three for almost every penalised home (98.1% and 98.6%), as it should for a housing model; what the scaffold removes is about half of the penalty carried by it. In this design rule C1 gives LIME fooled, KernelSHAP fooled and GeoShapley fooled, and rule C1′ gives LIME fully fooled, KernelSHAP partly fooled and GeoShapley partly fooled.

Explainer	Deployed model	Top 3, penalised homes (%)	Top 3, all homes (%)	Penalty share kept	Fidelity, all homes (%)	Fidelity, penalised homes (%)	Queries reaching f (%)	Detector AUC
LIME	Biased model, no scaffold	100.0 ± 0.0	67.3 ± 9.2	1.00 ± 0.00	–	–	–	–
	Scaffold, generic detector	100.0 ± 0.0	59.1 ± 12.0	0.96 ± 0.01	100.0 ± 0.0	100.0 ± 0.0	95.8 ± 1.2	0.96 ± 0.01
	Scaffold, matched detector (90%)	0.0 ± 0.0	0.1 ± 0.3	0.02 ± 0.02	95.9 ± 1.3	97.4 ± 4.6	0.0 ± 0.0	1.00 ± 0.00
KernelSHAP	Biased model, no scaffold	100.0 ± 0.0	86.3 ± 4.7	1.00 ± 0.00	–	–	–	–
	Scaffold, generic detector	97.0 ± 2.4	38.2 ± 14.0	0.52 ± 0.05	74.4 ± 1.3	73.8 ± 7.2	41.1 ± 2.1	0.73 ± 0.01
	Scaffold, matched detector (90%)	99.5 ± 1.4	51.7 ± 16.9	0.67 ± 0.07	89.5 ± 2.0	85.9 ± 11.2	53.6 ± 3.1	0.78 ± 0.01
GeoShapley	Biased model, no scaffold	100.0 ± 0.0	86.6 ± 4.5	1.00 ± 0.00	–	–	–	–
	Scaffold, generic detector	95.7 ± 4.1	41.6 ± 12.9	0.54 ± 0.06	74.4 ± 1.3	73.8 ± 7.2	45.0 ± 2.5	0.70 ± 0.01
	Scaffold, matched detector (90%)	100.0 ± 0.0	56.6 ± 16.1	0.72 ± 0.07	89.7 ± 2.1	87.9 ± 9.8	60.8 ± 3.2	0.72 ± 0.01

Table 2: Feature design: the penalty is triggered by the tract share column. Mean ± 95% interval over seeds; 200 explained homes per seed. Penalty share kept = signed attribution to the trigger on penalised homes divided by the same under the biased model served directly. Top 3 = share of explained homes whose trigger attribution (in absolute value) is among the three largest, on the penalised homes and on all 200. Fidelity = share of the explained (real) homes the scaffold serves with f; on penalised homes it is the share that still receive the penalty. AUC = probability that a real home scores higher than an explainer query on the detector.

5.3 What hiding costs: the stealth–fidelity frontier

An attacker who wants to hide more from a Shapley explainer has one lever: withhold the bias from more real homes. Lowering the matched threshold so that 75% and then 50% of real homes receive f lowers KernelSHAP's share kept from 0.67 to 0.47 and 0.24, and GeoShapley's from 0.72 to 0.50 and 0.29 (Table 4, Figure 3). The share kept falls roughly in step with fidelity. Even at 50% fidelity — a scaffold that charges only 45.8% (KernelSHAP) and 52.2% (GeoShapley) of the homes it was built to penalise — the trigger is still in the top three for 54.7% and 58.0% of penalised homes.

The generic detector, trained on different perturbations and run at a fixed threshold, reaches 74.4% fidelity and a share kept of 0.52 (KernelSHAP) and 0.54 (GeoShapley). Paired by seed against the matched curve at the same fidelity, it keeps +0.05 (interval −0.03 to +0.14) and +0.04 (−0.06 to +0.14) more: in the feature design a detector built without the explainer's queries lands on, or slightly above, the matched detector's curve, so the trade-off is set mainly by where the explainer queries the model. In the location design the generic detector sits above the curve (+0.07 and +0.08), significantly so for GeoShapley ($p = 0.036$, uncorrected): there a matched detector buys the attacker more. The net share falls to 0.17 and 0.14 at 75% fidelity and below zero at 50% (−0.15 and −0.17, intervals −0.23 to −0.07 and −0.27 to −0.07): at that threshold the explanations show less location effect on penalised homes than the honest model's do.

5.4 GeoShapley's spatial design protects it little, and its decomposition changes

Compared with KernelSHAP (rule C2). Paired by seed, GeoShapley minus KernelSHAP in the share kept under the matched scaffold is +0.05 (interval +0.02 to +0.08) in the feature design and 0.00 (−0.05 to +0.04) in the location design. By rule C2: feature design, GeoShapley slightly harder to fool; location design, no detectable difference at this sample size. The difference follows where the two explainers query. GeoShapley's queries are harder to tell from real homes (difference in detector AUC −0.05 in the feature design and −0.03 in the location design), consistent with its moving both coordinates together so that every hybrid sits at a real location, and more of them reach f at the same fidelity (+5.0 percentage points in the feature design). The routing curves show the same thing size by size (Figure 5). Treating location as a joint player helps only as far as it

Explainer	Deployed model	Top 3, penalised homes (%)	Top 3, all homes (%)	Penalty share kept	Share kept, net of honest	Fidelity, all homes (%)	Fidelity, penalised homes (%)	Queries reaching f (%)	Detector AUC
LIME	Honest model	68.6 ± 15.8	66.1 ± 10.9	0.57 ± 0.08	0.00 ± 0.00	–	–	–	–
	Biased model, no scaffold	93.9 ± 6.6	94.6 ± 5.0	1.00 ± 0.00	1.00 ± 0.00	–	–	–	–
	Scaffold, generic detector	88.9 ± 12.0	90.3 ± 9.2	0.96 ± 0.03	0.90 ± 0.07	100.0 ± 0.0	100.0 ± 0.0	94.9 ± 3.2	0.97 ± 0.01
	Scaffold, matched detector (90%)	1.6 ± 4.4	0.8 ± 1.3	0.05 ± 0.01	−1.24 ± 0.38	96.9 ± 1.5	98.9 ± 2.9	0.0 ± 0.0	1.00 ± 0.00
KernelSHAP	Honest model	97.1 ± 6.0	96.7 ± 1.7	0.33 ± 0.03	0.00 ± 0.00	–	–	–	–
	Biased model, no scaffold	100.0 ± 0.0	98.7 ± 1.4	1.00 ± 0.00	1.00 ± 0.00	–	–	–	–
	Scaffold, generic detector	97.5 ± 1.9	87.6 ± 3.4	0.46 ± 0.03	0.20 ± 0.05	71.9 ± 2.7	65.2 ± 5.3	42.0 ± 2.2	0.71 ± 0.01
	Scaffold, matched detector (90%)	98.1 ± 5.4	95.8 ± 3.7	0.65 ± 0.09	0.48 ± 0.15	90.0 ± 4.0	84.7 ± 16.5	59.3 ± 3.1	0.74 ± 0.02
GeoShapley	Honest model	92.6 ± 5.8	90.8 ± 3.2	0.32 ± 0.03	0.00 ± 0.00	–	–	–	–
	Biased model, no scaffold	100.0 ± 0.0	92.9 ± 4.7	1.00 ± 0.00	1.00 ± 0.00	–	–	–	–
	Scaffold, generic detector	96.7 ± 4.2	68.1 ± 5.2	0.45 ± 0.03	0.18 ± 0.05	71.9 ± 2.7	65.2 ± 5.3	43.4 ± 2.2	0.69 ± 0.01
	Scaffold, matched detector (90%)	98.6 ± 2.4	84.2 ± 8.1	0.65 ± 0.07	0.48 ± 0.13	88.8 ± 4.0	82.1 ± 10.5	63.6 ± 3.1	0.70 ± 0.01

Table 3: Location design: the penalty is decided inside the model from the coordinates; the trigger is location (both UTM columns for LIME and KernelSHAP; for GeoShapley the location player, which carries the intrinsic GEO effect plus half of each feature-by-location interaction). Mean ± 95% interval over seeds; 200 explained homes per seed. Penalty share kept = signed attribution to the trigger on penalised homes divided by the same under the biased model served directly; the net share first subtracts what the honest model attributes to location on those homes, so it is negative when an explanation shows less location effect than even the honest model does. Top 3 = share of explained homes whose trigger attribution (in absolute value) is among the three largest, on the penalised homes and on all 200. Fidelity = share of the explained (real) homes the scaffold serves with f; on penalised homes it is the share that still receive the penalty. AUC = probability that a real home scores higher than an explainer query on the detector.

moves the explainer's queries towards real homes, and the gain is small next to what the scaffold still hides.

The decomposition moves (rule C3). Under the matched scaffold the intrinsic location effect on penalised homes moves towards zero and the feature-by-location interactions grow (Table 5, Figure 4). In the feature design GEO on penalised homes goes from −0.11 under the biased model to 0.02 under the matched scaffold, and the per-home sum of absolute interaction effects from 0.07 to 0.16. Here GEO never carried the penalty, which sits on the trigger column; the change is the legitimate location effect fading, because hybrids routed to ψ, which ignores location, dilute it. In the location design GEO goes from −0.35 (honest model: −0.11) to −0.16, and the interactions from 0.07 to 0.14. Each change has a paired interval that excludes zero. The scaffold's routing depends on how real a hybrid looks rather than on any single feature, and GeoShapley reports the resulting discontinuities as features acting differently at different locations. In the location design an auditor who reads only the GEO map therefore sees less of the penalty than the location player as a whole carries.

5.5 Why: efficiency and coalition sizes

Efficiency. Under the matched scaffold the Shapley explanations of penalised homes still carry 0.79 (KernelSHAP) and 0.83 (GeoShapley) of the penalty gap the biased model's explanation puts between penalised and unpenalised homes: 0.67 and 0.72 on the trigger and 0.12 and 0.11 moved to other features, with grade the largest recipient in 4 of 5 seeds for KernelSHAP. Because the background term in the efficiency identity is the same for every home, the rest can leave the explanation only through explained homes that the scaffold itself serves with ψ. LIME's explanation carries −0.01 of the gap. The penalty that efficiency keeps in a Shapley explanation can be moved

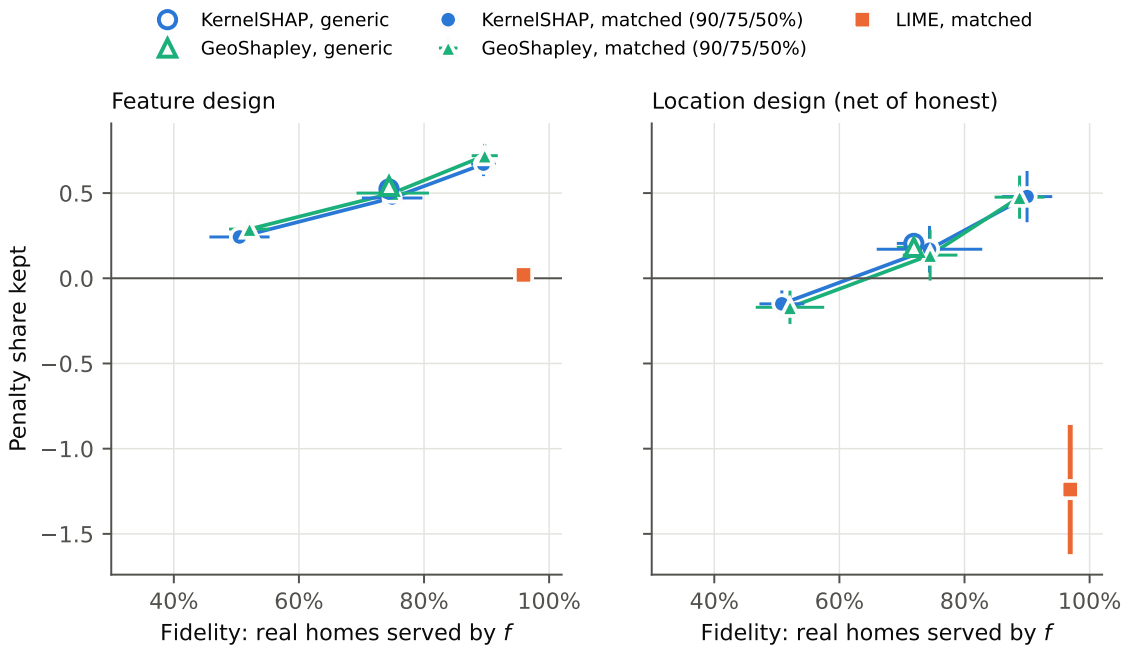


Figure 3: The stealth–fidelity frontier. Each filled marker is the matched detector at a stricter threshold (90%, 75% and 50% fidelity on the calibration homes); hollow markers are the generic detector at 0.5. The location panel shows the share kept net of the honest model. Mean and 95% interval over seeds in both directions.

but not removed while the explained home is served by f.

Routing by coalition size. Figure 5 shows the share of KernelSHAP's hybrids routed to f at each coalition size. It is U-shaped: 0.87 for a background home with only the trigger swapped in, 0.86 for the explained home itself ($k = 11$), and 0.48 at its lowest ($k = 6$). Both ends of the coalition range are near-real homes the detector cannot reject without rejecting real homes; only the middle is far enough from the data to be routed away, and equation (11) gives every size the same weight. Stricter thresholds lower the whole curve, the ends included, which is why hiding costs fidelity.

Decomposition check (rule M) and a simple law. The independent Monte Carlo evaluation of equation (11) reproduces the measured share kept with a mean absolute error of 0.002 for KernelSHAP (seed-to-seed standard deviation of the measured share 0.037; prediction holds), 0.003 for GeoShapley in the feature design (prediction holds) and 0.002 in the location design, raw share (prediction holds; Table 6). Since it evaluates the same Shapley average with the same detector, this mainly confirms the decomposition and our implementation. The informative comparison is with the plain average of the routing rates over sizes, which ignores how the penalty is spread across sizes: it is within 0.019 of the measured share for KernelSHAP, 0.026 for GeoShapley and 0.040 in the location design (descriptive, not pre-specified). To a first approximation, *the share of the penalty a Shapley explainer keeps is the average, over coalition sizes, of the share of its hybrids the detector routes to the biased model.*

Design	Explainer	Detector, threshold	Fidelity, all homes (%)	Fidelity, penalised homes (%)	Penalty share kept	Top 3, penalised homes (%)	Queries reaching f (%)
Feature	KernelSHAP	Matched, τ^*	89.5 ± 2.0	85.9 ± 11.2	0.67 ± 0.07	99.5 ± 1.4	53.6 ± 3.1
		Matched, 75%	74.9 ± 4.8	72.9 ± 9.4	0.47 ± 0.05	91.1 ± 6.9	33.8 ± 3.7
		Matched, 50%	50.5 ± 4.8	45.8 ± 7.6	0.24 ± 0.02	54.7 ± 12.3	15.6 ± 2.1
		Generic, 0.5	74.4 ± 1.3	73.8 ± 7.2	0.52 ± 0.05	97.0 ± 2.4	41.1 ± 2.1
	GeoShapley	Matched, τ^*	89.7 ± 2.1	87.9 ± 9.8	0.72 ± 0.07	100.0 ± 0.0	60.8 ± 3.2
		Matched, 75%	75.0 ± 5.8	72.2 ± 9.1	0.50 ± 0.03	93.8 ± 2.3	40.4 ± 3.3
		Matched, 50%	52.1 ± 3.3	52.2 ± 9.7	0.29 ± 0.03	58.0 ± 9.5	22.0 ± 2.0
		Generic, 0.5	74.4 ± 1.3	73.8 ± 7.2	0.54 ± 0.06	95.7 ± 4.1	45.0 ± 2.5
Location	KernelSHAP	Matched, τ^*	90.0 ± 4.0	84.7 ± 16.5	0.48 ± 0.15	98.1 ± 5.4	59.3 ± 3.1
		Matched, 75%	74.4 ± 8.4	67.9 ± 12.7	0.17 ± 0.14	89.8 ± 6.8	38.4 ± 4.1
		Matched, 50%	50.8 ± 3.6	42.2 ± 7.1	−0.15 ± 0.08	63.3 ± 11.2	20.8 ± 2.6
		Generic, 0.5	71.9 ± 2.7	65.2 ± 5.3	0.20 ± 0.05	97.5 ± 1.9	42.0 ± 2.2
	GeoShapley	Matched, τ^*	88.8 ± 4.0	82.1 ± 10.5	0.48 ± 0.13	98.6 ± 2.4	63.6 ± 3.1
		Matched, 75%	74.5 ± 4.4	65.1 ± 13.2	0.14 ± 0.15	88.4 ± 6.7	43.2 ± 4.4
		Matched, 50%	52.1 ± 5.5	45.2 ± 6.1	−0.17 ± 0.10	56.8 ± 11.0	24.3 ± 3.7
		Generic, 0.5	71.9 ± 2.7	65.2 ± 5.3	0.18 ± 0.05	96.7 ± 4.2	43.4 ± 2.2

Table 4: The stealth–fidelity frontier for the Shapley explainers. Each matched row uses the same detector with a stricter threshold. Penalty share kept is net of the honest model in the location design. Mean ± 95% interval over seeds.

5.6 Robustness

Model families and audit settings. Table 7 and Figure 6 repeat the feature-design attack with random forests (honest R^2 0.86) and neural networks (0.81) in place of XGBoost, and against auditors who change the explainer's settings after the attacker has trained its detector. In 13 of 14 explainer–configuration cells the C1′ class is the same as in the main result (rule R); exceptions: LIME Gaussian sampler (LIME: partly fooled). With random forests the matched scaffold leaves LIME 0.02 of the penalty and KernelSHAP and GeoShapley 0.67 and 0.72 with neural networks 0.02, 0.67 and 0.72. The model family barely matters, for a structural reason. The explainers' queries do not depend on the model, so the detector, its threshold and its routing are the same for every family (seed by seed, the fidelities and the queries reaching f are identical). Because neither h nor ψ uses the trigger, the model enters the trigger's attribution only through $(r_{S\cup j} - r_S)\,(h - \psi)$, where r_S indicates that hybrid (x_S, b) is routed to f: only hybrids whose routing changes when the trigger is swapped in contribute. The penalty term, $-\delta\,(r_{S\cup j}\mathbf{1}_{S\cup j} - r_S\mathbf{1}_S)$ with $\mathbf{1}_S$ indicating that the hybrid is penalised, is model-free. Changing the Shapley background moves the share kept by a few hundredths at most, and more LIME samples change nothing. The exception is LIME's Gaussian sampler, discussed in Section 6: its queries are not the ones the detector was trained on, 31.0% of them reach f, and LIME keeps 0.37 of the penalty.

Spatial autocorrelation of the trigger (rule C5). Breaking up the spatial structure of the trigger makes the scaffold worse at hiding, not better. As Moran's I of the trigger falls from 0.83 to 0.22 and 0.00, the share of KernelSHAP's queries that reach f goes from 53.6% to 64.9% and 63.7%, and its share kept from 0.67 to 0.74 and 0.75; GeoShapley moves the same way (0.72 to 0.77; Table 8, Figure 7). Most of the change comes between the observed and the intermediate level. The slope of the matched leak on Moran's I is −0.14 (interval −0.21 to −0.06) for KernelSHAP and −0.11 (−0.12 to −0.10) for GeoShapley: by rule C5, KernelSHAP, leak depends on Moran's I; GeoShapley, leak

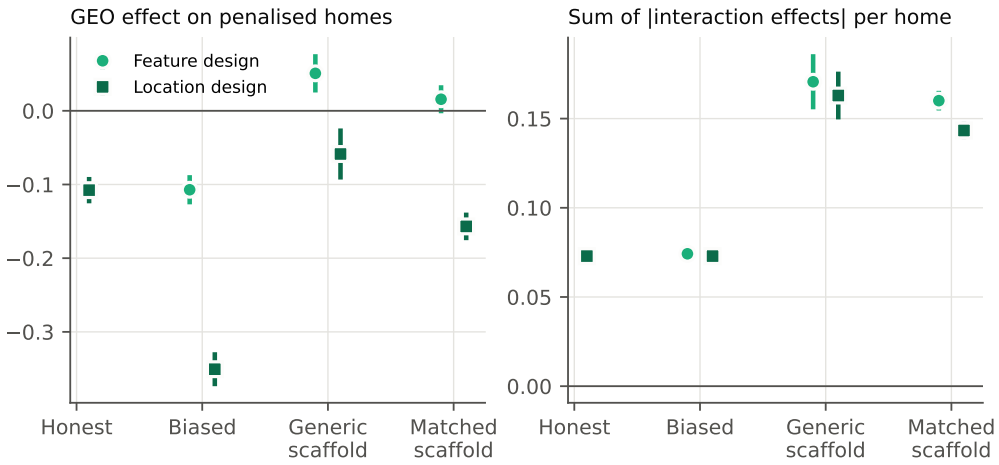


Figure 4: GeoShapley's decomposition by deployment: the intrinsic location effect (GEO) on penalised homes, and the per-home sum of absolute feature-by-location interaction effects. Mean and 95% interval over seeds.

Design	Deployed model	GEO on penalised homes	Mean |GEO|	Sum of |interaction effects|
Feature	Biased model, no scaffold	-0.11 ± 0.02	0.12 ± 0.01	0.07 ± 0.00
	Scaffold, generic detector	0.05 ± 0.03	0.06 ± 0.01	0.17 ± 0.02
	Scaffold, matched detector (90%)	0.02 ± 0.02	0.06 ± 0.01	0.16 ± 0.01
Location	Honest model	-0.11 ± 0.02	0.12 ± 0.01	0.07 ± 0.00
	Biased model, no scaffold	-0.35 ± 0.02	0.17 ± 0.01	0.07 ± 0.00
	Scaffold, generic detector	-0.06 ± 0.03	0.07 ± 0.02	0.16 ± 0.01
	Scaffold, matched detector (90%)	-0.16 ± 0.02	0.09 ± 0.01	0.14 ± 0.00

Table 5: GeoShapley's decomposition by deployment. GEO is the intrinsic location effect; interaction effects are the feature-by-location terms, summed in absolute value per home and averaged. Mean $\pm$ 95% interval over seeds.

depends on Moran's I; LIME, whose queries are rejected at every level, no detectable dependence. A reading consistent with this: a strongly autocorrelated trigger is predictable from location, so a hybrid that pairs one home's tract share with another home's coordinates is internally inconsistent and the detector can reject it; when the trigger is spatially random, such hybrids look like real homes and reach f. Spatial segregation of the protected attribute thus works for the attacker: the more clustered it is, the more a scaffold can hide from a Shapley explainer at the same fidelity.

A smaller penalty. At $\delta = 0.10$, a 21% price cut, LIME is still fully fooled (share kept 0.03), while KernelSHAP and GeoShapley keep a larger share than at $\delta = 0.30$: 0.80 and 0.85 at the same fidelity, and 0.32 and 0.39 at 50% fidelity (Table 9). The detector and its routing do not depend on δ (seed by seed the fidelities and the queries reaching f are identical to the main run), so the penalty term of the trigger's attribution scales with δ. The term $(r_{S\cup j} - r_S)(h - \psi)$ does not, and it is a larger share of a smaller penalty. Because a smaller penalty is also a smaller attribution, the trigger is less prominent to begin with: it is in the top three for 93.3% of penalised homes without a scaffold (both Shapley explainers) and 68.1% (KernelSHAP) and 74.0% (GeoShapley) under the

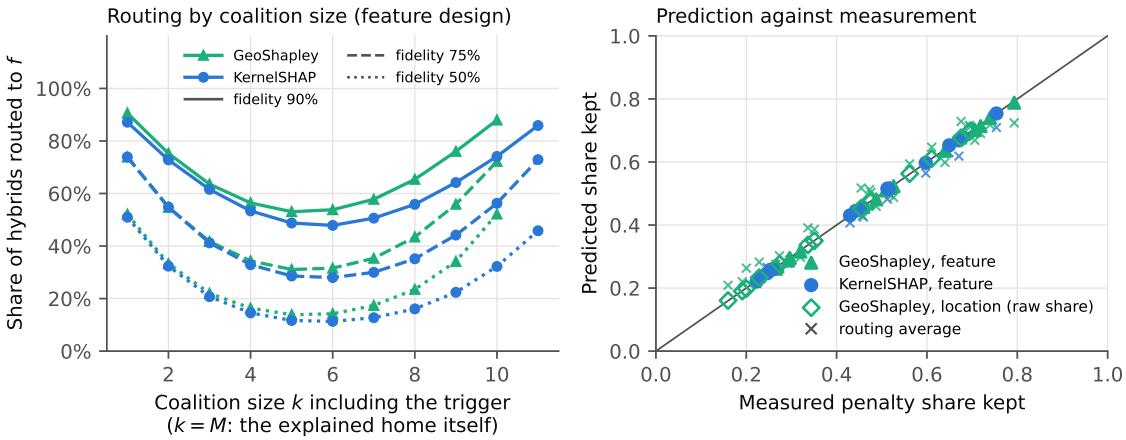


Figure 5: Mechanism. Left: share of with-trigger hybrids routed to f by coalition size, at the three attacker thresholds (feature design). Right: predicted against measured penalty share kept, one point per seed and threshold; crosses are the plain average of the routing rates.

matched scaffold.

5.7 An outcome-based check sees what the explainers miss

No explainer is needed to see this penalty. On the 600 held-out homes, the 8-nearest-neighbour mean of the deployed model's residuals separates penalised from unpenalised homes with AUC 0.95 at $\delta = 0.30$ and 0.81 (interval 0.77–0.85) at $\delta = 0.10$, for the scaffold that fools LIME completely (Table 10, Figure 8; 10 seeds). The AUC uses the true penalised set, so it measures how well the residual surface separates the groups, not what an auditor would find unaided. The label-free step, LISA low–low clusters, flags 29.0% of the penalised homes at $\delta = 0.10$, and 64.7% of the homes it flags are penalised. It raises few false alarms under the honest model, flagging 3.9% of its unpenalised homes. The check loses power for penalties of a few percent (AUC 0.52 at $\delta = 0.02$). The scaffold cannot evade it without changing what real homes are charged, because it is computed from served predictions and observed prices only. By rule E7 (go: include section) the check is reported here, although the point estimate clears the 0.80 threshold narrowly and its interval includes values below it. It is a complement to explanation-based audits, not a defence we evaluate against adaptive attackers.

6 Discussion

The estimator, not the geography. What separates the explainers in our results is where each one queries the model. LIME's quartile sampler produces rows that no real home resembles; a detector separates them perfectly, and LIME then explains the innocuous model while the deployed model keeps penalising almost every real home. The Shapley explainers are harder to fool for two reasons that have nothing to do with space. Efficiency ties their attributions to the served prediction at the explained home, so the penalty cannot leave the explanation while that home is served by the biased model; and their coalition hybrids include, at both ends of the size range, points that differ from a real home in one or two values and that a detector cannot reject without rejecting real homes. GeoShapley moves the two coordinates together, which keeps every hybrid at a real location. Its queries are harder to detect, more of its hybrids reach the biased model at every coalition size

Design, explainer	Attacker fidelity target	Fidelity (%)	Predicted share kept	Routing average	Measured share kept	Seeds
Feature, GeoShapley	90%	89.7 ± 2.1	0.72 ± 0.07	0.68 ± 0.06	0.72 ± 0.07	5
	75%	75.0 ± 5.8	0.50 ± 0.03	0.47 ± 0.04	0.50 ± 0.03	5
	50%	52.1 ± 3.3	0.29 ± 0.03	0.28 ± 0.02	0.29 ± 0.03	5
Feature, KernelSHAP	90%	89.5 ± 2.0	0.67 ± 0.07	0.64 ± 0.07	0.67 ± 0.07	5
	75%	74.9 ± 4.8	0.47 ± 0.05	0.45 ± 0.05	0.47 ± 0.05	5
	50%	50.5 ± 4.8	0.24 ± 0.02	0.25 ± 0.03	0.24 ± 0.02	5
Location, GeoShapley (raw share)	90%	88.8 ± 4.0	0.65 ± 0.07	0.68 ± 0.07	0.65 ± 0.07	5
	75%	74.5 ± 4.4	0.42 ± 0.09	0.46 ± 0.08	0.42 ± 0.09	5
	50%	52.1 ± 5.5	0.21 ± 0.05	0.26 ± 0.05	0.21 ± 0.05	5

Table 6: Mechanism check on the penalised explained homes. *Predicted*: an independent Monte Carlo evaluation of Equation (11) for the scaffold, divided by the same for the biased model, with the attack's detector and thresholds and the auditor's background rows (200 hybrids per home and coalition size; pre-specified, rule M). *Routing average*: the plain mean over coalition sizes of the share of with-trigger hybrids the detector routes to f (descriptive, not pre-specified). *Measured*: the penalty share kept in the attack run of the same seed; in the location design the raw share (not net of the honest model), which is what the routing decomposition predicts. Because *Predicted* evaluates the same Shapley average with the same detector, its agreement with *Measured* checks the decomposition and the implementation; the routing average is the substantive approximation. Mean ± 95% interval over seeds.

(Figure 5), and by rule C2 in the feature design GeoShapley keeps slightly more of the penalty than KernelSHAP, while in the location design we find no detectable difference between GeoShapley and KernelSHAP. Its other players are still recombined across homes, so its middle coalitions are still largely routed away. Treating location carefully is valuable for interpretation; as a defence it helps only as far as it moves the explainer's queries towards real data, and here that is a few points of the penalty. The question to ask of any explainer used in an audit is the one the routing curves answer: what share of its queries looks like real data?

Where geography does enter: segregation helps the attacker. Space matters to the attack through the data, not through the explainer. The more clustered the protected attribute, the more a scaffold hides from a Shapley explainer at the same fidelity (rule C5). In a segregated city a tract's demographic share is close to a function of location, so hybrids that break that relation are easy to spot and route away; when the attribute is spatially random they look like real homes. Within the range we tested, this suggests that Shapley-based audits are weakest where spatial bias is most likely: in highly segregated places.

What partial hiding means for an audit. "Partly fooled" is not "safe"; nor does it mean the penalty has left the explanations. Against the matched scaffold KernelSHAP and GeoShapley keep between about half and three quarters of the penalty, and on the penalised homes themselves the trigger stays among the three largest attributions. An auditor who disaggregates explanations by the protected attribute, or reads the explanations of homes in the affected tracts, would still see it. What the scaffold removes is what a summary audit reads: the size of the effect, and the trigger's rank in global importance, which falls because the scaffold also shrinks the trigger's attribution on the unpenalised homes. An attacker who accepts lower fidelity can do more: at 50% fidelity the

Configuration	Explainer	Penalty share kept	Fidelity (%)	Top 3, all homes (%)	Top 3, penalised homes (%)	Seeds	Same class as main
Main (XGBoost, default settings)	LIME	0.02 ± 0.02	95.9 ± 1.3	0.1 ± 0.3	0.0 ± 0.0	5	yes
	KernelSHAP	0.67 ± 0.07	89.5 ± 2.0	51.7 ± 16.9	99.5 ± 1.4	5	yes
	GeoShapley	0.72 ± 0.07	89.7 ± 2.1	56.6 ± 16.1	100.0 ± 0.0	5	yes
Random forest	LIME	0.02 ± 0.01	95.9 ± 1.3	0.2 ± 0.3	0.7 ± 1.9	5	yes
	KernelSHAP	0.67 ± 0.07	89.5 ± 2.0	55.1 ± 17.8	100.0 ± 0.0	5	yes
	GeoShapley	0.72 ± 0.07	89.7 ± 2.1	58.9 ± 17.7	100.0 ± 0.0	5	yes
Neural net	LIME	0.02 ± 0.01	95.9 ± 1.3	0.2 ± 0.3	0.7 ± 1.9	5	yes
	KernelSHAP	0.67 ± 0.07	89.5 ± 2.0	50.6 ± 17.2	98.7 ± 2.3	5	yes
	GeoShapley	0.72 ± 0.07	89.7 ± 2.1	55.5 ± 14.1	99.2 ± 2.1	5	yes
Background 25 rows	KernelSHAP	0.67 ± 0.07	89.5 ± 2.0	58.3 ± 13.5	98.3 ± 3.0	5	yes
	GeoShapley	0.72 ± 0.06	89.7 ± 2.1	60.0 ± 12.2	98.8 ± 2.0	5	yes
Background 200 rows	KernelSHAP	0.67 ± 0.07	89.5 ± 2.0	50.3 ± 8.6	99.5 ± 1.4	5	yes
	GeoShapley	0.72 ± 0.07	89.7 ± 2.1	53.4 ± 5.3	100.0 ± 0.0	5	yes
k-means background	KernelSHAP	0.64 ± 0.07	89.5 ± 2.0	39.8 ± 4.2	96.8 ± 4.9	5	yes
	GeoShapley	0.68 ± 0.07	89.7 ± 2.1	45.6 ± 9.0	99.4 ± 1.8	5	yes
LIME 5,000 samples	LIME	0.00 ± 0.01	95.9 ± 1.3	0.0 ± 0.0	0.0 ± 0.0	5	yes
LIME Gaussian sampler	LIME	0.37 ± 0.03	95.9 ± 1.3	17.2 ± 6.8	50.1 ± 18.7	5	**no**

Table 7: Robustness of the matched scaffold, feature design. Audit-setting rows keep the attacker's detector trained for the default settings and change only what the auditor does; the biased reference uses the same changed settings. Class = fully fooled (trigger in top three $\leq 5\%$, share kept ≤ 0.10, fidelity $\geq 90\%$, as fixed before the runs) or partly fooled. Configurations with fewer than three complete seeds are not shown.

scaffold still charges 45.8% of the targeted homes the full penalty while KernelSHAP shows 0.24 of it. The frontier in Figure 3 turns this into a quantity a regulator can reason about: for a given explainer, how much bias can be hidden at what cost in bias actually delivered, which is the fidelity on penalised homes reported in Tables 2–4.

GeoShapley under attack. For geographers the more specific lesson is about how GeoShapley fails. Under the scaffold the intrinsic location effect on penalised homes shrinks towards zero and the feature-by-location interactions roughly double (rule C3). In the location design, where location carries the penalty, a GEO map read on its own would understate it: the GEO effect alone keeps 0.20 of the excess penalty on penalised homes, against 0.48 for the location player as a whole. In both designs the interaction terms would invite a substantive story — features that matter differently in different places — that is in fact an artefact of the attacker's routing. We did not set out to build a detector and do not evaluate one, but inflated interactions without a corresponding change in the model's fit are a pattern worth examining in future work.

Reconciling with earlier results. Slack et al. (2020) found LIME easier to fool than SHAP on tabular benchmarks; we find the same ordering in a housing market. An earlier version of this project reported the opposite, that LIME resists the attack in a geographic setting. That result came from the generic detector, which is trained on Gaussian noise and lets 95.8% of LIME's real queries through. Replications of scaffold attacks should match the detector to the explainer's actual sampler, or report how many queries reach the biased model.

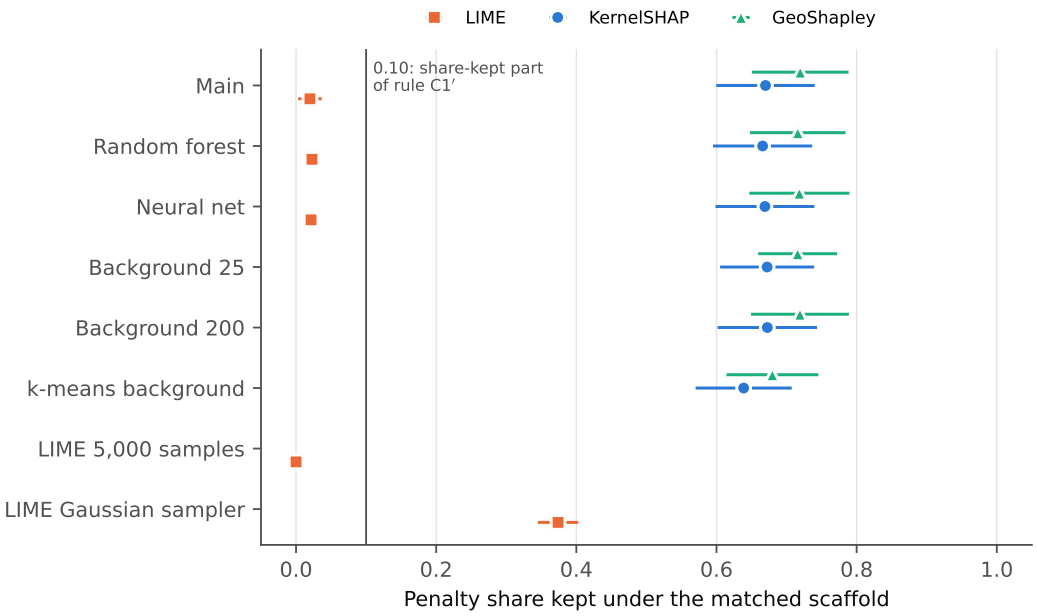


Figure 6: Penalty share kept under the matched scaffold across model families and audit settings the attacker did not train for (feature design). Mean and 95% interval over seeds.

Changing the audit settings. An auditor who changes the Shapley background after the attacker has trained its detector gains little: with 25 rows, 200 rows or 100 k-means centroids KernelSHAP keeps 0.67, 0.67 and 0.64 of the penalty (default 0.67), because any background still produces hybrids of real homes whose ends look real. LIME is different. More samples change nothing (0.00), but switching to its Gaussian sampler, whose queries the quartile-trained detector has not seen, lets 31.0% of LIME's queries reach the biased model, and LIME then shows 0.37 of the penalty: the one audit setting in which an explainer's class changes (rule R). This fits the main result. What protects an explainer is a query distribution the attacker did not anticipate, not its treatment of space, and an attacker who expects the auditor to switch samplers can train on both, which we did not test.

Looking at outcomes. The penalty that the scaffold hides from explainers is plainly visible in the residuals of the model's own predictions: their local mean separates penalised homes with AUC 0.81 even at a 21% penalty, using tools — local means, Moran's I, LISA — that are standard in geography. Without labels, LISA's low–low clusters point to about 29.0% of the penalised homes at that penalty: enough to send an auditor to the right places, not a list of the affected homes. The check needs observed outcomes, loses power for penalties of a few percent, and we have not tested it against an attacker who anticipates it. It complements explanation-based audits rather than replacing them, and it reflects a broader point: for spatial bias, an audit that looks at what a model does to places can be more robust than one that asks the model to explain itself.

7 Limitations

- One metropolitan area in this draft. [**TODO:** Cook County runs for the submission version]
- The penalty is synthetic and large; Table 9 repeats the main result at a smaller one.

Trigger Moran's I	Explainer	Generic: queries reaching f (%)	Matched: queries reaching f (%)	Matched: penalty share kept	Seeds
0.83	LIME	95.8 ± 1.2	0.0 ± 0.0	0.02 ± 0.02	5
	KernelSHAP	41.1 ± 2.1	53.6 ± 3.1	0.67 ± 0.07	5
	GeoShapley	45.0 ± 2.5	60.8 ± 3.2	0.72 ± 0.07	5
0.22	LIME	96.2 ± 1.6	0.0 ± 0.0	0.01 ± 0.00	5
	KernelSHAP	47.3 ± 2.2	64.9 ± 2.4	0.74 ± 0.03	5
	GeoShapley	50.6 ± 2.5	70.4 ± 3.4	0.76 ± 0.06	5
0.00	LIME	95.9 ± 1.9	0.0 ± 0.0	0.01 ± 0.01	5
	KernelSHAP	53.6 ± 2.4	63.7 ± 4.3	0.75 ± 0.06	5
	GeoShapley	56.5 ± 2.4	68.9 ± 3.2	0.77 ± 0.05	5

Table 8: Spatial autocorrelation of the trigger, lowered by giving a share of homes the trigger value of a random other home (seeds 0–4 at every level). Mean ± 95% interval.

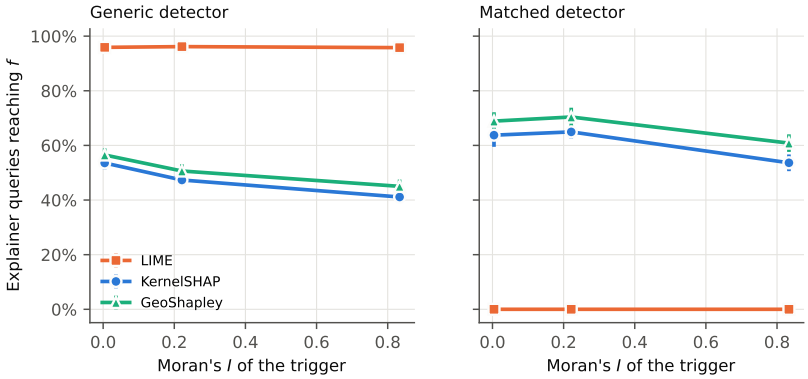


Figure 7: Queries reaching f under the matched scaffold as the trigger's spatial autocorrelation is lowered. Mean and 95% interval over seeds.

- The attacker knows which explainer is used; Table 7 varies the settings it does not know.
- The adversary is static: defences and attackers that adapt to them are out of scope. We also did not vary the attacker's detector (its model class or the number of harvested queries). A stronger detector could lower the share kept at a given fidelity, so our shares describe one reasonable attacker, not the strongest possible.
- The auditor's Shapley background is drawn from the training homes, which the matched detector saw as real during training; near-empty coalitions may look real partly because the detector has memorised those rows. A k-means background, whose rows no real home matches exactly, changes the share kept little (Table 7), but a background from homes the attacker never saw would be the cleaner test.
- 200 explained homes per seed, of which about 32 are penalised; the number of seeds is in every table.
- Feature-design GeoShapley uses the package's sampled coalitions, so its values carry Monte Carlo error that the seed intervals include.

Explainer	Deployed model	Top 3, penalised homes (%)	Penalty share kept	Fidelity (%)
LIME	Biased model, no scaffold	55.8 ± 17.0	1.00 ± 0.00	–
	Scaffold, matched detector (90%)	0.0 ± 0.0	0.03 ± 0.05	95.9 ± 1.3
KernelSHAP	Biased model, no scaffold	93.3 ± 6.0	1.00 ± 0.00	–
	Scaffold, matched detector (90%)	68.1 ± 6.8	0.80 ± 0.09	89.5 ± 2.0
	Matched detector, 75% fidelity	47.5 ± 7.4	0.59 ± 0.06	74.9 ± 4.8
	Matched detector, 50% fidelity	14.3 ± 9.8	0.32 ± 0.03	50.5 ± 4.8
GeoShapley	Biased model, no scaffold	93.3 ± 6.0	1.00 ± 0.00	–
	Scaffold, matched detector (90%)	74.0 ± 6.8	0.85 ± 0.09	89.7 ± 2.1
	Matched detector, 75% fidelity	53.1 ± 5.2	0.63 ± 0.04	75.0 ± 5.8
	Matched detector, 50% fidelity	25.8 ± 11.2	0.39 ± 0.03	52.1 ± 3.3

Table 9: Penalty 0.10 in $\log_{10}$ price (a 21% price cut) instead of 0.30, feature design, seeds 0–4.

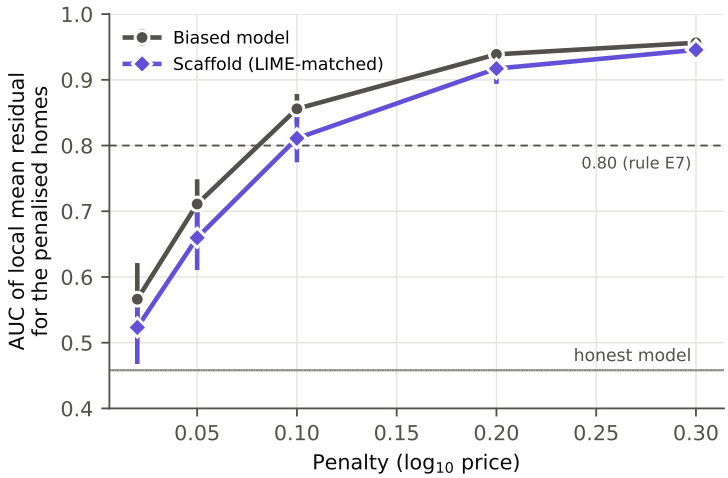


Figure 8: Outcome-based check: AUC of the local mean residual for the penalised homes, by penalty size, for the biased model and the LIME-matched scaffold; the line marks the honest model. Mean and 95% interval over seeds.

8 Conclusion

A scaffold can hide a racial price penalty from the three explainers most used in geographic machine learning. How much it can hide is set by each explainer's estimator: LIME, whose queries are unlike real homes, is fooled almost for free, while KernelSHAP and GeoShapley, bound by efficiency and querying many near-real hybrids, can be fooled only partly, and only at a cost in the bias actually delivered. Making an explainer spatially explicit matters only through where it queries the model: GeoShapley's joint location player keeps its hybrids at real locations and buys it a few points of the penalty, not protection. Geography matters through the data instead: the more segregated the protected attribute, the more a scaffold can hide. Under attack GeoShapley's location effect shrinks and spurious interactions appear, so its decomposition should not be read as evidence that a model treats places fairly. Audits of spatial models should ask where an explainer queries the model, should look at explanations of the affected places and not only at global summaries, and should pair explanations with checks on outcomes that a model cannot route around.

Model	Penalty	Moran's I of residuals	AUC, own residual	AUC, local mean residual	LISA low-low recall (%)	LISA flags, unpenalised (%)
Biased	0.02	0.04 ± 0.02	0.56 ± 0.03	0.57 ± 0.05	8.2 ± 4.1	3.6 ± 0.9
Biased	0.05	0.05 ± 0.02	0.66 ± 0.02	0.71 ± 0.04	17.6 ± 5.0	3.3 ± 0.7
Biased	0.10	0.12 ± 0.02	0.79 ± 0.02	0.86 ± 0.02	32.3 ± 5.8	2.8 ± 0.4
Biased	0.20	0.27 ± 0.02	0.94 ± 0.01	0.94 ± 0.01	60.3 ± 4.2	2.5 ± 0.3
Biased	0.30	0.38 ± 0.02	0.98 ± 0.00	0.96 ± 0.00	70.4 ± 2.8	2.2 ± 0.5
Honest	0.00	0.03 ± 0.02	0.48 ± 0.02	0.46 ± 0.06	4.9 ± 2.5	3.9 ± 0.9
Scaffold (LIME-matched)	0.02	0.04 ± 0.02	0.54 ± 0.03	0.52 ± 0.06	6.8 ± 3.1	3.5 ± 1.2
Scaffold (LIME-matched)	0.05	0.05 ± 0.02	0.64 ± 0.03	0.66 ± 0.05	14.5 ± 4.0	3.0 ± 0.8
Scaffold (LIME-matched)	0.10	0.09 ± 0.02	0.76 ± 0.02	0.81 ± 0.04	29.0 ± 6.1	2.9 ± 0.5
Scaffold (LIME-matched)	0.20	0.22 ± 0.03	0.91 ± 0.02	0.92 ± 0.02	53.7 ± 5.7	2.4 ± 0.3
Scaffold (LIME-matched)	0.30	0.33 ± 0.03	0.95 ± 0.01	0.95 ± 0.01	64.1 ± 5.4	2.1 ± 0.3

Table 10: Outcome-based check on the 600 held-out homes, no explainer involved: residual = served prediction minus observed $\log_{10}$ price; 8 nearest neighbours; AUC against the true penalised set; LISA at conditional-permutation $p < 0.05$. Mean ± 95% interval over seeds.

Data and code availability

Housing sample: `seattle_sample_3k.csv` from the GeoShapley repository. Demographics: ACS 2011–2015 table B03002 and 2015 TIGER/Line tracts, US Census Bureau. Code: the scripts listed in Appendix C, run with fixed seeds; repository `geospatial-xai-attacks`, branch `paper-draft-v1`.

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Design	Seed	LIME K	LIME fid.	KernelSHAP K	KernelSHAP fid.	GeoShapley K	GeoShapley fid.
Feature	0	0.04	96.0	0.75	89.5	0.79	90.5
	1	0.01	96.0	0.65	91.5	0.74	91.5
	2	0.02	97.5	0.67	87.5	0.71	89.5
	3	0.00	94.5	0.68	90.5	0.72	90.0
	4	0.03	95.5	0.60	88.5	0.64	87.0
Location	0	−1.31	96.0	0.47	92.5	0.42	87.0
	1	−1.40	97.0	0.56	92.5	0.55	90.0
	2	−0.78	99.0	0.57	88.5	0.55	88.5
	3	−1.14	96.0	0.52	91.5	0.54	93.5
	4	−1.58	96.5	0.27	85.0	0.32	85.0

Table 11: Per-seed penalty share kept (K, net of honest in the location design) and fidelity (%) under the matched scaffold.

Design	Detector	Threshold	Fidelity, explained (%)	Fidelity, calibration (%)	r(scaffold, f), test homes	Synthetic training rows
Feature	lime_gen	0.50 ± 0.00	100.0 ± 0.0	99.8 ± 0.4	1.00 ± 0.00	–
	lime_mat	0.50 ± 0.00	95.9 ± 1.3	95.5 ± 1.4	0.99 ± 0.01	72000 ± 0
	shap_gen	0.50 ± 0.00	74.4 ± 1.3	75.4 ± 2.3	0.94 ± 0.01	–
	kernelshap_mat	0.21 ± 0.02	89.5 ± 2.0	90.8 ± 0.7	0.97 ± 0.01	22923 ± 4538
	geoshapley_mat	0.17 ± 0.01	89.7 ± 2.1	90.5 ± 0.4	0.97 ± 0.00	36238 ± 211
Location	lime_gen	0.50 ± 0.00	100.0 ± 0.0	99.9 ± 0.2	1.00 ± 0.00	–
	lime_mat	0.50 ± 0.00	96.9 ± 1.5	96.2 ± 0.9	0.99 ± 0.01	72000 ± 0
	shap_gen	0.50 ± 0.00	71.9 ± 2.7	72.2 ± 2.8	0.92 ± 0.01	–
	kernelshap_mat	0.27 ± 0.02	90.0 ± 4.0	90.8 ± 0.8	0.97 ± 0.01	11366 ± 2204
	geoshapley_mat	0.17 ± 0.01	88.8 ± 4.0	90.5 ± 0.5	0.97 ± 0.01	30511 ± 324

Table 12: Detector diagnostics. Generic detectors at threshold 0.5; matched detectors at τ^*. Fidelity on the 200 explained homes (not used to set τ^*) and on the 400 calibration homes; r is the correlation of scaffold and biased predictions on all 600 test homes.

A Per-seed results

B Detector diagnostics and the design without coordinates

Design without coordinates. Table 13 repeats the feature design with the coordinates removed from every model (LIME and KernelSHAP only; GeoShapley needs coordinates). LIME is again fully fooled (share kept 0.02). KernelSHAP keeps 0.74, more than with coordinates (0.67), and 66.0% of its queries reach f against 53.6%. This fits the autocorrelation result (rule C5, Results): without coordinates the detector cannot reject hybrids whose tract share is inconsistent with their location. With two fewer players the coalition structure also changes, so the comparison is indicative only.

C Reproducibility

Code. All in `scripts/`:

- `run_matched_attack.py`: one design and seed per call; resumable; every detector and every

Explainer	Deployed model	Top 3, penalised homes (%)	Top 3, all homes (%)	Penalty share kept	Fidelity, all homes (%)	Fidelity, penalised homes (%)	Queries reaching f (%)	Detector AUC
LIME	Biased model, no scaffold	100.0 ± 0.0	54.7 ± 9.2	1.00 ± 0.00	–	–	–	–
	Scaffold, generic detector	100.0 ± 0.0	51.7 ± 10.1	0.99 ± 0.02	100.0 ± 0.0	100.0 ± 0.0	98.8 ± 2.1	0.94 ± 0.01
	Scaffold, matched detector (90%)	0.0 ± 0.0	0.4 ± 0.8	0.02 ± 0.01	96.6 ± 1.8	97.9 ± 4.2	0.0 ± 0.0	1.00 ± 0.00
KernelSHAP	Biased model, no scaffold	100.0 ± 0.0	80.2 ± 12.8	1.00 ± 0.00	–	–	–	–
	Scaffold, generic detector	98.2 ± 2.2	52.4 ± 16.4	0.61 ± 0.02	76.5 ± 3.7	78.1 ± 5.9	50.1 ± 2.5	0.68 ± 0.02
	Scaffold, matched detector (90%)	100.0 ± 0.0	60.6 ± 18.6	0.74 ± 0.04	90.2 ± 2.0	88.5 ± 5.8	66.0 ± 2.0	0.70 ± 0.01

Table 13: Design without coordinates (LIME and KernelSHAP only). Mean ± 95% interval over seeds; 200 explained homes per seed. Penalty share kept = signed attribution to the trigger on penalised homes divided by the same under the biased model served directly. Top 3 = share of explained homes whose trigger attribution (in absolute value) is among the three largest, on the penalised homes and on all 200. Fidelity = share of the explained (real) homes the scaffold serves with f; on penalised homes it is the share that still receive the penalty. AUC = probability that a real home scores higher than an explainer query on the detector.

block of 25 explained homes is cached; all writes are atomic;

- `summarize_matched_attack.py`: seed-level tables;
- `mechanism_routing.py`: Section 4.6;
- `run_residual_diagnostic.py`: the outcome check;
- `fidelity_by_group.py`: fidelity on penalised and unpenalised explained homes, from the cached detectors;
- `make_paper_tables.py` and `make_paper_figures_matched.py`: every table, figure and number in this draft;
- `verify_paper_numbers.py`: an independent recomputation of the main numbers and rule outcomes from the raw attribution files and cached detectors, compared with the macros and the compiled PDF.

Only complete runs enter a table or a rule; a configuration enters the robustness table with at least three complete seeds. The job lists that were run are in `docs/paper/data/queues_run.txt`.

Environment. Python 3.10.20 on x86_64, numpy 2.2.6, scikit-learn 1.7.2, XGBoost 3.2.0, shap 0.49.1, lime 0.2.0.1, geoshapley 0.2.1.1. Before any run we reproduced the reference run at seed 42, made earlier on a Linux aarch64 machine with the same package versions: honest R^2, innocuous R^2, LIME query routing under both detectors, and KernelSHAP's trigger attribution for the biased model and the matched scaffold agreed to at least twelve decimal places, and the matched KernelSHAP detector routed exactly the same share of queries. Mean compute per seed was about 36 minutes of explainer time on one core.

Changes during the audit of 12 November. (i) KernelSHAP is run with `l1_reg=False`; (ii) GeoShapley's coalition sampling is seeded; (iii) all seeds were recomputed from scratch with the final code, so no table mixes code versions or machines.

D Reconciliation with earlier project results

- **"LIME resists the attack in a geographic setting."** Withdrawn. The generic (Slack-recipe) LIME detector is trained on Gaussian noise in raw units and lets 95.8% of LIME's real queries through to f (Table 2); LIME then simply explains the biased model. With a detector matched to LIME's sampler, LIME is the most completely fooled of the three explainers, the ordering Slack et al. (2020) report.
- **"GeoShapley location attribution falls by 93%."** Withdrawn. It was computed with a retired custom GeoSHAP and divided an adversarial GeoSHAP value by an honest KernelSHAP one; the October summary already corrected it to −58% under the generic detector.
- **"A 26% price penalty."** Corrected: prices are on a base-10 log scale, so the 0.30 penalty halves the predicted price and the 0.10 penalty cuts it by 21%.
- **16 October attack-settings table.** Its "SHAP-matched" detector reached high hiding only by routing 7.5% of real homes to f; the threshold rule here fixes fidelity at 90% or more before hiding is measured.
- **Draft v0 (synthetic stand-in for the trigger, one seed).** Its KernelSHAP and GeoShapley figures (trigger in the top three for 13% and 16% of homes under the matched scaffold) came from a synthetic trigger column and a single seed; on the real column the matched scaffold leaves the trigger in the top three for 51.7% and 56.6% of all explained homes and 99.5% and 100.0% of the penalised ones (Table 2).